MACHINE LEARNING PROJECT

Predictive Customer Churn Classifier(Random Forest and XGBoost)

A structured record of the project, stage by stage.

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DATASET
Telco-Customer-Churn.csv

1 Business Understanding

Chrun: telecom loses customers, predict leavers early so they retain the customers.

2 Data Understanding (EDA)

1. s your section 2, same content and structure, just with the typos and spelling fixed:

    1. The dataset contains **7,043 rows and 21 columns**, representing 7,043 observations and 21 variables.
    2. **dtypes:** float64(1), int64(2), object(18)
    3. **Missing values per column:** 0 (null)
    4. **Target balance:** we checked how many No and Yes and found: No = 5,174 customers stayed (73.5%), Yes = 1,869 left (26.5%).
    5. **Visual part of EDA:**
        1. **Overall churn balance:** the chart shows more than 5,000, which means 5,174 customers stayed, and on the other side the graph shows under 2,000, which means around 1,869 people left.
        2. **Churn by contract type:**
            - Month-to-month: the chart shows 42% of customers churned (left) and 58% stayed.
            - One-year contract: 10% churned, while 90% stayed.
            - Two-year contract: only approximately 2% churned, while 98% stayed.
        3. **Tenure by churn:**
            - Pattern 1: high churn among newer customers.
            - Pattern 2: churn counts tend to decrease as tenure increases.
            - Pattern 3: many long-term customers remain.
        4. **Monthly charges by churn:**
            - Observation 1: customers who churned have a higher median monthly charge. The median for customers who left appears to be around $79, compared with around $64 for customers who stayed.
        5. **Churn rate by monthly charges and tenure:**
            - Customers with short tenure have higher churn rates.
            - Customers in higher monthly-charge groups have higher churn rates within the same tenure group.
        6. **Distribution of monthly charges:**
            - There is a larger concentration of customers around $20.
            - Many customers pay between approximately $70 and $105.
            - Customers have a wide range of monthly packages.

3 Data Preparation

1. `Fix TotalCharges: it's text, with 11 blanks for brand-new customers`
2. `Drop the ID column (useless for prediction)`
3. `Encode target: Yes=1 (churned), No=0 (stayed)`
4. `One-hot encode all remaining categorical columns`

4 Train / Test Split

Split the data 80/20 into training and test sets using `train_test_split`. Used `stratify=y` to keep the churn ratio (about 27%) identical in both sets, and `random_state=42` so the split is reproducible.

Training set: 5,634 customers. Test set: 1,409 customers. Churn rate in both: about 0.265.

The test set was held out before any balancing, so it stays a realistic sample and prevents data leakage.

5 Handle Class Imbalance

The target is imbalanced: about 73% of customers stayed and only 27% churned. Training on this directly would bias the model toward predicting "no churn," so it would miss the customers who actually leave.

Applied SMOTE (Synthetic Minority Over-sampling Technique) to balance the classes. SMOTE creates synthetic churner examples until both classes are equal.

Applied SMOTE to the **training set only**, never the test set. Balancing the test set would leak synthetic data into evaluation and give misleading scores.

Before SMOTE: 4,139 stayed, 1,495 churned. After SMOTE: 4,139 each. Training set grew from 5,634 to 8,278 rows.

6 Modeling

Trained two ensemble models on the SMOTE-balanced training data:

- Random Forest (200 trees, default depth)
- XGBoost (300 trees, learning rate 0.1, max depth 5)

Both were tested on the untouched real test set. Random Forest was chosen as the final model.

7 Evaluation

Used ROC-AUC as the main metric, since accuracy is misleading on imbalanced data. Also checked precision, recall, F1, and a confusion matrix.

Random Forest: ROC-AUC 0.821, churn recall 0.60, churn precision 0.58.
XGBoost: ROC-AUC 0.819, churn recall 0.60, churn precision 0.56.

Both exceeded the 0.80 target. The model catches about 60% of real churners. Recall was prioritised over precision, since a retention offer costs far less than losing a customer.

8 Interpretation

Top churn drivers (Random Forest feature importance): TotalCharges, tenure, MonthlyCharges, electronic-check payment, fiber-optic internet, and contract type.

These match the EDA patterns, which builds confidence in the model. I also ran a feature experiment: the model reaches near-full performance with only about 15 of 30 features, showing the trade-off between simplicity and accuracy.

9 Communication and Next Steps

This project predicts which telecom customers are likely to leave, so the company can act before they go. The final Random Forest model scores ROC-AUC 0.82 and catches about 60% of churners in the test set.

Business recommendations:

1. Target new customers in their first year, the highest-risk window.
2. Move month-to-month customers onto annual contracts with incentives.
3. Investigate the electronic-check and fiber-optic segments for service or pricing issues.
4. Focus retention budget on high-charge, low-tenure customers.

Limitations: churn depends on factors not in the data (competitor offers, customer service experience), so 0.82 is a realistic ceiling here. The model shows correlation, not proof of cause.

Next steps: tune the decision threshold to catch more churners if the business wants higher recall, test the lean 15-feature model in production for speed, and retrain regularly as new data arrives.